Danny Quang

Email: dquang@ucsd.edu
Mobile: 925-577-8678

Github: github.com/dquangucsd
Portfolio: dquangucsd.github.io/dq-portfolio

Education

- **University of California, San Diego** San Diego, California
  MS - Computer Science, BS - Computer Science, GPA: 3.53 *Expected Graduation: 2026*

Work Experience

- **Software Engineering DevOps Intern at Synaptics**: June-September 2025 C, Python, Embedded C
  - Maintain Continuous Integration (CI) computer systems and test fixture for Bluetooth Low Energy firmware products
  - Develop automated CI scripts for building, testing and reporting of functional testing
  - Develop and maintain test harness for automated system testing of wireless audio products
- **Software Engineering Intern at Wind River**: June-September 2023 C, Python, Embedded C
  - Ported company's proprietary Real-Time Operating system (VxWorks7) to the Raspberry Pi Zero 2W
  - Designed a pinmux driver from scratch for the GPIO for general purpose ability and various other parts
  - VxWorks7 on the Pi Zero 2W - a commercial low power system - enables customers to run low cost, low power tasks
- **Tutor for UCSD CSE: CSE 110, CSE 29**: March-June 2023, March-June 2024 Javascript, CSS, HTML, C
  - Tutor for CSE 110, Software Engineering: focuses on building a project with front end technologies (HTML, CSS, JS) from the ground up in teams. Teach teams to employ AGILE practices, develop CI/CD pipelines (linter, etc.), and ownership of their roles as backend, Q&A, frontend, etc.
  - Tutor for CSE 29, Intro to Computer Systems: Lead Logistics Team. Focus on systems programming in C, software tools.
  - Attend weekly Supervision/ASE(s) and staff meetings, manage and organize logistics, perform individual and group tutoring, conduct labs, content review and debugging, and Slack support
- **Data Analyst Intern at The Center for Community Energy**: June-October 2022 Python, BeautifulSoup4, HTML
  - Web scraped the market involving V2G data, including multiple car company websites targeting key words
- **Webmaster for American Institute of Aeronautics and Astronautics**: May 2022-November 2023 Javascript, CSS
  - Redesigned, maintained, and updated the website for elected positions and events for AIAA UCSD

Projects

- **Privacy Airtag**: C
  - Developed firmware for STM32 (ARM Cortex M4) board to act like an Airtag. Use Bluetooth Low Energy (BLE) driver to periodically broadcast signals to receiver when stationary for over 1 minute, turn off when moving to protect privacy
  - Detected movement with LSM6DSL accelerometer, wake when there is significant, successive movement
  - Minimized power consumption by 40% by turning off all components when possible except for the accelerometer, and only waking necessary components on significant interrupt events
- **Quadcopter**: C, C++, Arduino, Fusion360
  - Designed schematic and laid out board in Fusion 360, and developed firmware and proportional-integral-derivative (PID) tuning for flight with Arduino
  - Designed firmware to reliably send and receive signals for control (using PID controller) and feedback (filtered inertial measurement unit (IMU) signal)
- **Twitter Clone**: Rust
  - Developed a distributed, load-balanced, scalable, fault-tolerant Twitter clone
  - Connected front end requests to the backend servers with clients using consistent hashing, formatted RPC calls
  - Implemented a keeper to for logical time and sync between the servers (inspired by Frangipani's Petal)
  - Used the async trait to meet performance requirements, like signing up 100 users concurrently within 5 seconds
- **UC Social Den**: React.js, Express.js, CSS, HTML, Javascript
  - Designed a meetup app in React with Node.js that allows UC San Diego students to post and join casual events
  - Used MongoDB and Amazon's S3 to store user data and profile images, along with Google's OAuth2 for login

Skills

- **Languages**: Python, Java, C, C++, Assembly, R, Swift, HTML, JavaScript, CSS, Node.js, React, SQL, Golang, Rust, Haskell, React.js, MongoDB
- **Frameworks**: ScikitLearn, Scipy, Matplotlib, Numpy, Pandas, Tensorflow (Keras), RegEx
- **Tools**: LaTeX, MATLAB, Vim, GitHub, Jupyter Notebook, VSCode, IntelliJ, UNIX, Git, JUnit, AWS EC2

Relevant Coursework: *Graduate Computer Networks, Computer Security, Computer Vision, Databases, Operating Systems, Compilers, Computer Architecture, Advanced Data Structures and Algorithms, Software Engineering, Multivariate Statistical Learning, Representation Learning, Programming Languages, Web Client Languages, Robotic System Design and Implementation, Distributed Systems, Differentiable Programming, Parallel Programming*